

Winter in Data Science 5.0
End-to-End Machine Learning Systems
Plant Disease Classification

Atharva Shetwe

Roll No: 24B4537

2nd Year Undergraduate

Department of IEOR

Abstract

This report presents a comprehensive end-to-end machine learning system for plant disease classification developed as part of Winter in Data Science 5.0. Using the PlantVillage dataset, the project progresses through exploratory data analysis, baseline classical machine learning models, deep learning approaches with convolutional neural networks, transfer learning, federated learning, and system-level considerations such as model persistence and visualization.

The study highlights the limitations of shallow models on high-dimensional image data, demonstrates the effectiveness of pretrained convolutional architectures for fine-grained disease recognition, and exposes the challenges introduced by decentralized training under federated learning. Beyond accuracy, the report emphasizes system robustness, data distribution effects, and operational considerations, reflecting a shift from experimentation toward deployable machine learning systems.

1 Introduction

Plant disease identification is a critical problem in agriculture, as delayed or incorrect diagnosis can result in significant crop losses. Automated disease recognition using computer vision and machine learning offers a scalable alternative to manual inspection, especially in regions with limited access to agricultural experts.

This project frames plant disease classification as an end-to-end machine learning systems problem rather than a purely modeling task. The focus extends beyond achieving high accuracy to understanding dataset characteristics, modeling limitations, training dynamics, and deployment-oriented considerations such as decentralization and observability.

The PlantVillage dataset serves as a controlled benchmark for studying these aspects. While its clean acquisition conditions enable strong performance, they also highlight the gap between benchmark accuracy and real-world deployment.

2 Dataset Description

The PlantVillage dataset is organized as a folder-based image dataset, where each directory corresponds to a unique Plant–Disease combination. In total, the dataset contains 54,305 color images spanning 38 disease classes across 14 distinct plant species.

For this study, only the color images were used to avoid redundancy introduced by grayscale and segmented variants. The dataset exhibits significant class imbalance, both globally and within individual plants. Tomato alone contributes over 18,000 images across multiple disease categories, while certain healthy classes contain fewer than 200 samples.

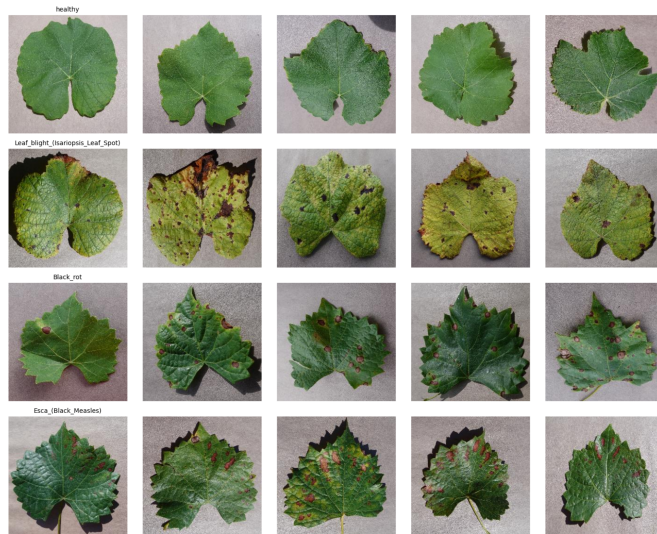


Figure 1: Representative leaf image from the PlantVillage dataset illustrating visible disease patterns under controlled imaging conditions.

The controlled acquisition setup results in high-quality images with minimal background clutter. However, variations in lighting, exposure, and subtle texture differences remain, making the task a fine-grained classification problem even under ideal conditions.

3 Exploratory Data Analysis

Exploratory data analysis was conducted to understand class distribution, plant-wise imbalance, and visual characteristics of the dataset.

3.1 Class and Plant Distribution

The dataset contains 38 plant-disease classes with highly uneven sample counts. Certain diseases such as citrus greening and tomato yellow leaf curl virus dominate the dataset, while others have relatively limited representation.

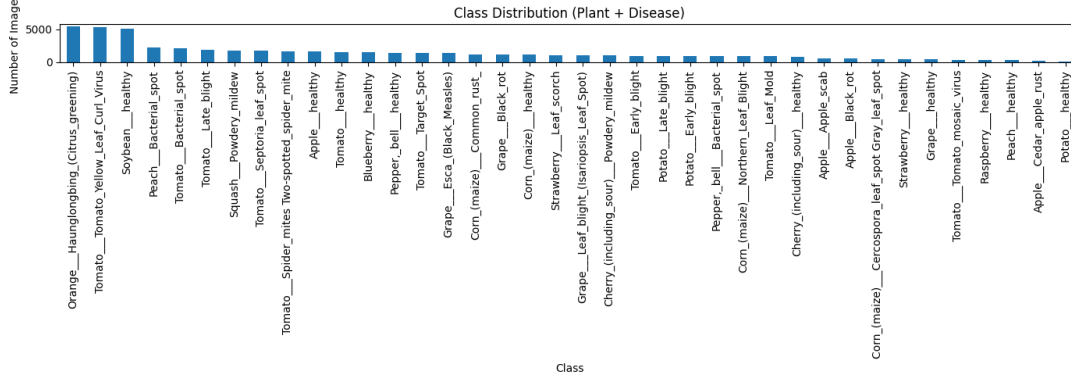


Figure 2: Distribution of images across plant-disease classes.

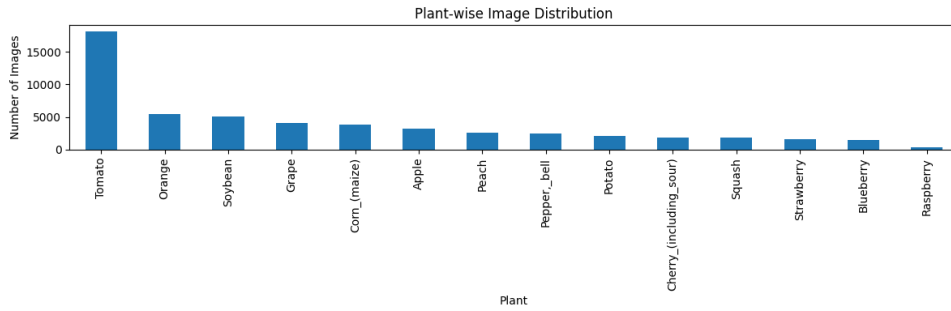


Figure 3: Plant-wise distribution of images in the dataset.

This imbalance suggests that naive training may bias models toward high-support classes, motivating careful evaluation and the use of robust feature extractors.

3.2 Visual Variability

Images within the same plant but different disease states often exhibit subtle differences in color intensity, spot morphology, and texture. These variations are challenging to capture using global statistics alone, reinforcing the need for spatially aware models such as CNNs.

4 Baseline Modeling with Classical Machine Learning

To establish performance baselines, classical machine learning models were trained on flattened image representations.

A Dummy Classifier using the most frequent class achieved approximately 10% accuracy, confirming correct preprocessing and serving as a lower bound.

A Random Forest classifier trained on flattened 64×64 RGB images achieved approximately 64% accuracy. While this represents a substantial improvement, performance varied widely across classes.

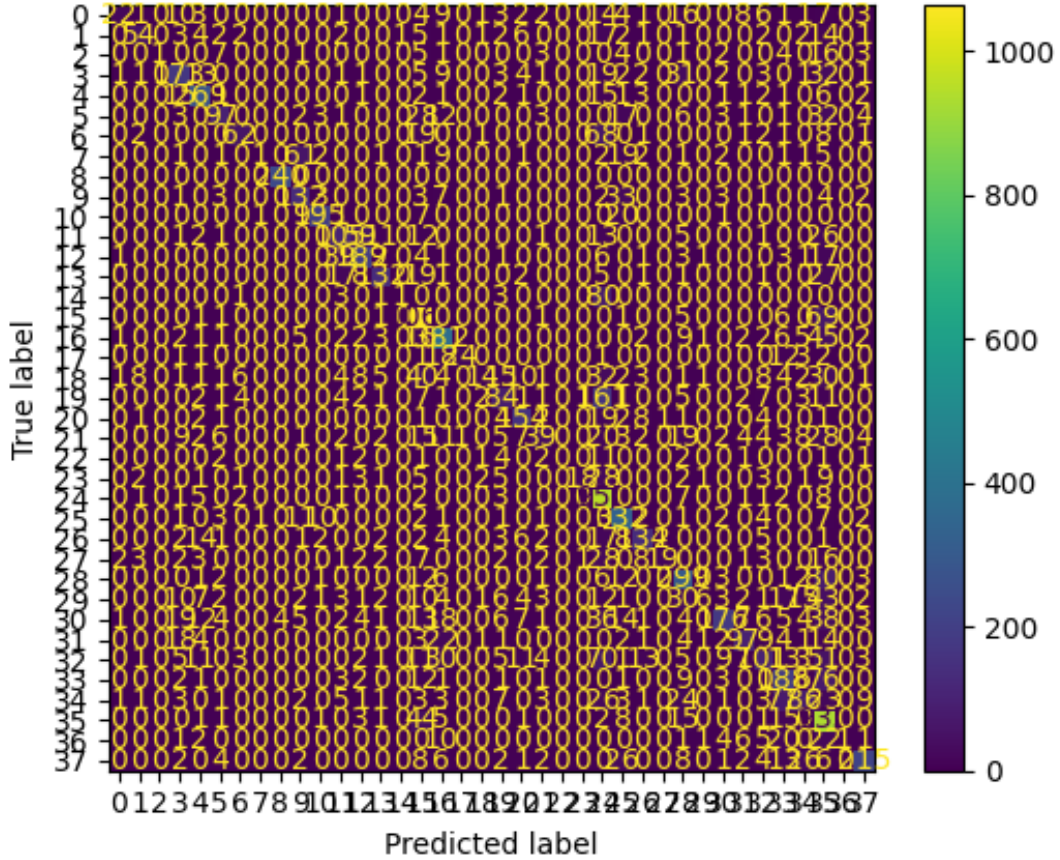


Figure 4: Confusion matrix for the Random Forest baseline model.

The confusion matrix reveals poor recall for minority classes and visually similar diseases. Flattened representations discard spatial structure, limiting the model’s ability to capture fine-grained disease patterns.

5 Deep Learning Approaches

5.1 Custom CNN

A custom convolutional neural network trained from scratch achieved high training accuracy but exhibited a noticeable generalization gap. Performance was strong for visually distinct diseases but weaker for subtle and minority classes.

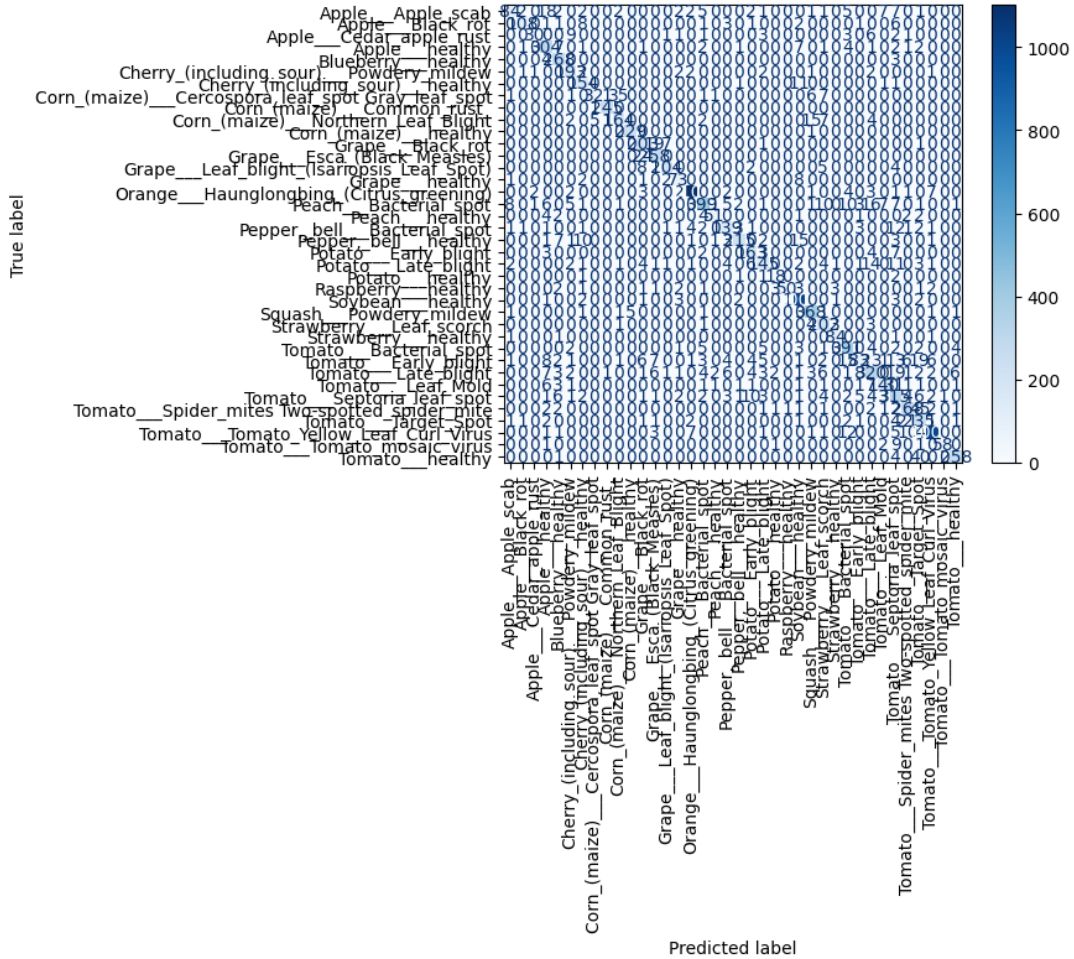


Figure 5: Confusion matrix for the custom CNN model.

This behavior highlights the risk of overfitting when training deep models on imbalanced datasets without pretrained representations.

5.2 Transfer Learning with MobileNetV2

Transfer learning using a pretrained MobileNetV2 backbone significantly improved performance across all metrics. Precision, recall, and F1-scores exceeded 0.95 for most classes after fine-tuning higher convolutional layers.

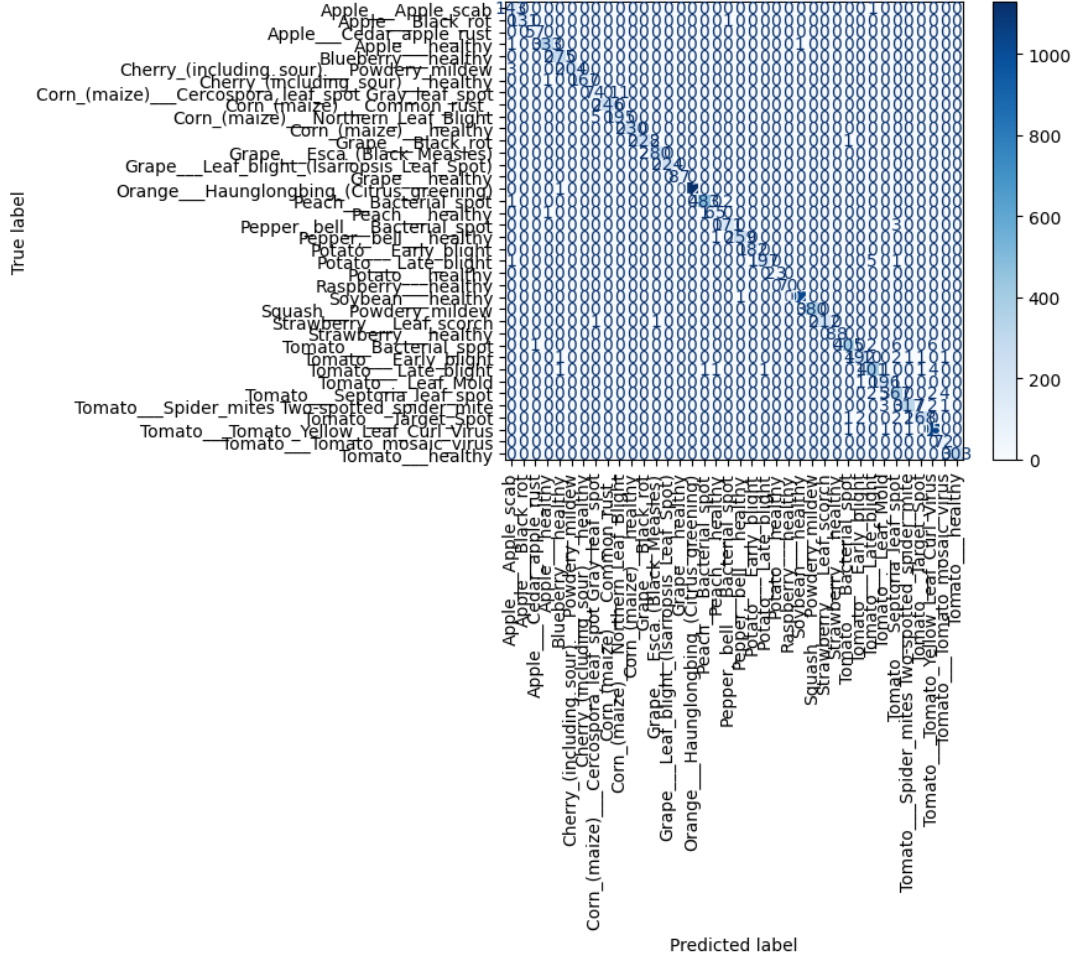


Figure 6: Confusion matrix for the MobileNetV2 transfer learning model.

Confusion between similar disease classes was substantially reduced, demonstrating the effectiveness of pretrained features for fine-grained plant disease recognition.

6 Federated Learning

The project was extended to a federated learning setting using the Flower framework. Multiple simulated clients trained locally on disjoint subsets of the dataset, with updates aggregated using the FedAvg algorithm.

The global model was initialized using centrally trained CNN weights. While initial federated rounds achieved strong accuracy, performance steadily degraded across rounds.

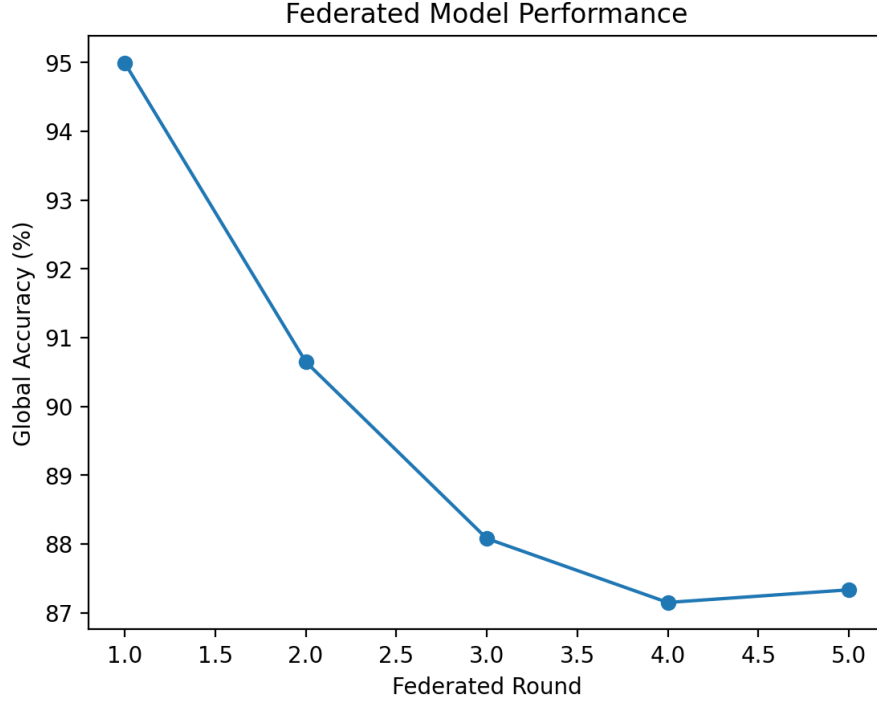


Figure 7: Global accuracy across federated learning rounds.

This degradation highlights the sensitivity of FedAvg to non-IID data distributions. Client drift toward local optima and instability in large fully connected layers contributed to performance loss, exposing challenges not present in centralized training.

7 Model Persistence, Visualization, and MLOps

Rather than focusing on further accuracy improvements, the final phase treated models as production artifacts. The final global federated model was saved to disk, and per-round metrics were logged for offline analysis.

A lightweight Streamlit application was developed to visualize global accuracy across federated rounds. Separating training from monitoring mirrors real-world MLOps workflows, enabling observability and debugging without retraining.

Persisting models and metrics ensures that even suboptimal training outcomes retain value by enabling diagnosis, comparison, and iterative improvement.

8 Conclusion

This project demonstrates that effective plant disease classification extends beyond model accuracy. Dataset characteristics, modeling choices, training paradigms, and system-level considerations all play critical roles.

While transfer learning achieves near-perfect performance on controlled benchmarks, federated learning exposes new challenges related to data heterogeneity and optimization stability. The transition from experimentation to engineering underscores the importance of robustness, interpretability, and deployability in real-world machine learning systems.

9 Learnings and Reflections

This project provided a comprehensive learning experience spanning data analysis, model development, distributed training, and system-level machine learning practices. Rather than focusing solely on predictive performance, the week-wise structure encouraged a deeper understanding of how different components of a machine learning pipeline interact and influence overall system behavior.

9.1 Data-Centric Insights

One of the most important learnings emerged during exploratory data analysis. Although the PlantVillage dataset is large, containing over 54,000 images, it is highly imbalanced across both plants and disease classes. The presence of 38 disease classes distributed across 14 plants resulted in uneven representation, with Tomato alone accounting for more than one-third of the dataset. This imbalance clarified that dataset size alone is insufficient for robust modeling; class distribution and intra-class variability play a critical role in determining model behavior.

The visual inspection of disease samples further reinforced that plant disease classification is a fine-grained recognition problem. Diseases within the same plant often differ only in subtle texture patterns, lesion boundaries, or color intensity. This observation explained why simple statistical or pixel-based methods struggle, and highlighted the necessity of models capable of learning spatial hierarchies.

9.2 Modeling Trade-offs

The progression from shallow models to deep learning provided clear quantitative and qualitative insights. The Dummy Classifier achieving approximately 10% accuracy served as a sanity check and reinforced the importance of establishing lower-bound baselines. The Random Forest baseline, achieving roughly 64% accuracy, demonstrated that classical machine learning models can capture coarse visual cues such as dominant colors and textures, but fail to generalize well to minority classes and visually similar diseases.

Training a convolutional neural network from scratch significantly improved performance, but also revealed overfitting risks. Despite achieving very high training accuracy, the custom CNN exhibited reduced recall for minority classes and subtle disease varia-

tions. This gap emphasized that model capacity alone does not guarantee generalization, especially under class imbalance.

Transfer learning with a pretrained MobileNetV2 backbone proved to be a turning point. Leveraging pretrained representations resulted in consistently high precision, recall, and F1-scores across nearly all classes. Fine-tuning higher-level layers further improved performance, reinforcing the insight that pretrained visual features provide a strong inductive bias for agricultural image recognition tasks.

9.3 Federated Learning Reflections

Extending the problem to a federated learning setting introduced a fundamentally different set of challenges. While the federated model achieved strong accuracy in the initial round due to centralized pretraining, performance degraded across subsequent rounds. This behavior highlighted the sensitivity of the FedAvg algorithm to non-IID data distributions.

Each simulated client contained different plant and disease compositions, causing local updates to drift toward client-specific optima. The naive averaging of these updates led to the gradual loss of globally useful features. Additionally, the large fully connected layers of the CNN amplified this instability, making the model particularly vulnerable to divergence under federated aggregation. This experience demonstrated that federated learning is not a drop-in replacement for centralized training, but a paradigm that requires careful architectural and algorithmic design.

9.4 System-Level and MLOps Learnings

The final phase of the project shifted focus from model accuracy to system robustness and usability. Saving trained models and logging per-round metrics transformed experimental outputs into reusable artifacts. The ability to visualize federated training dynamics using a Streamlit application made performance degradation immediately interpretable, reinforcing the importance of observability in distributed systems.

Separating training, evaluation, and visualization workflows mirrored real-world machine learning operations, where reproducibility and monitoring are as critical as model performance. This phase emphasized that even imperfect models retain value when their behavior is well-documented and analyzable.

9.5 Overall Reflection

Overall, this project demonstrated that effective machine learning systems are built through iterative refinement across multiple dimensions: data understanding, model selection, training strategy, and operational design. High accuracy achieved under con-

trolled conditions does not guarantee robustness in decentralized or real-world settings. By progressing from exploratory analysis to federated deployment considerations, the project reinforced the importance of adopting a holistic, systems-oriented mindset when developing machine learning solutions.